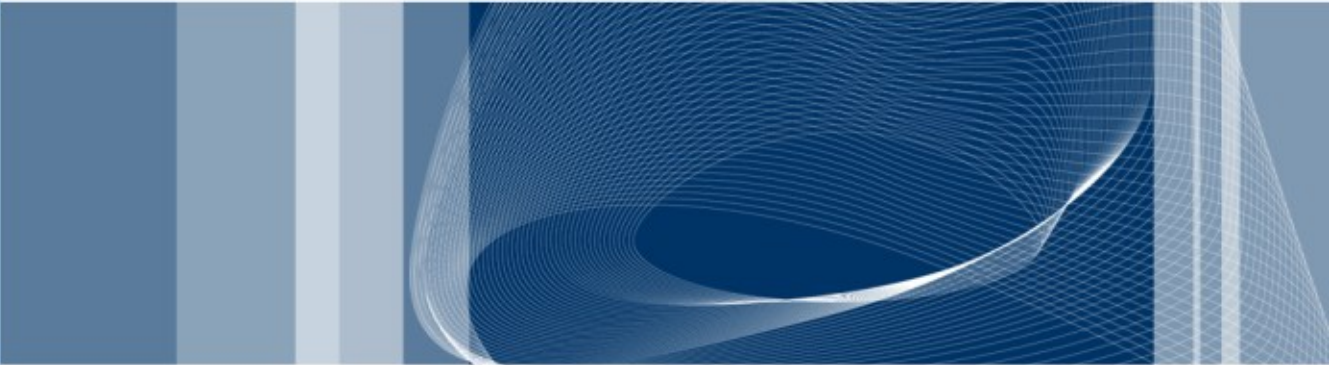




Master Degree – Mechanical engineering
A.A: 2023-2024
Modeling of Mechanical Behaviour of Materials and Finite
Element Simulation

**MODELING OF A SOFT PNEUMATIC ACTUATOR USING AN
HYPERELASTIC MATERIAL**

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Analysed structure

WHAT ARE WE GOING TO ANALYSE?

A unidirectional soft pneumatic actuator, also known as «soft robot»

Assumption for the model

→ Uniform 3D solid

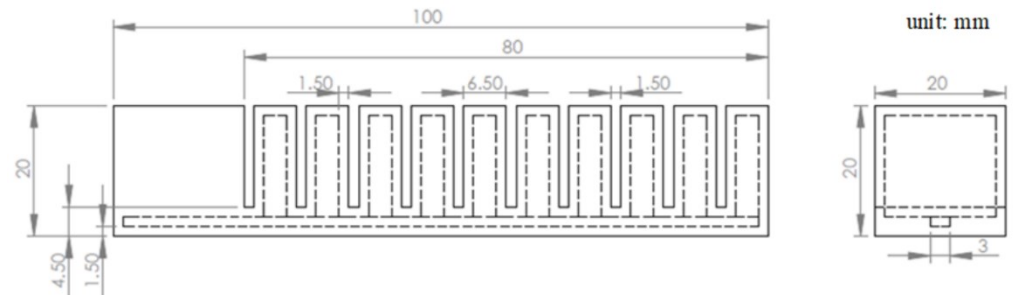
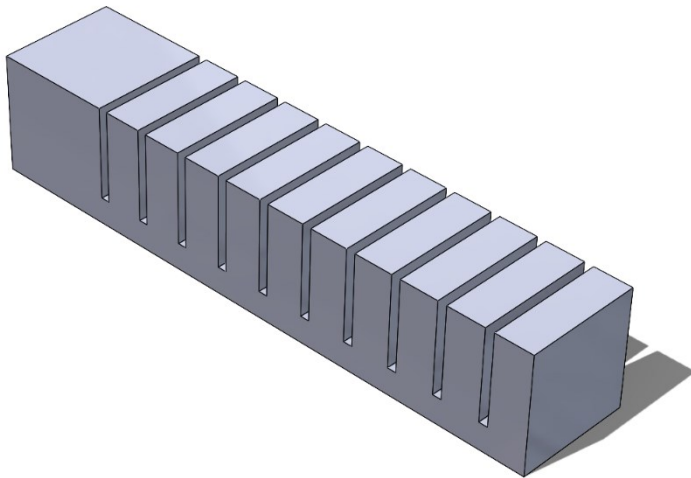
→ No gravity involved

Model's characteristics

→ Non linear almost incompressible elastic material

→ Large displacements and strains

→ Contact problem involved

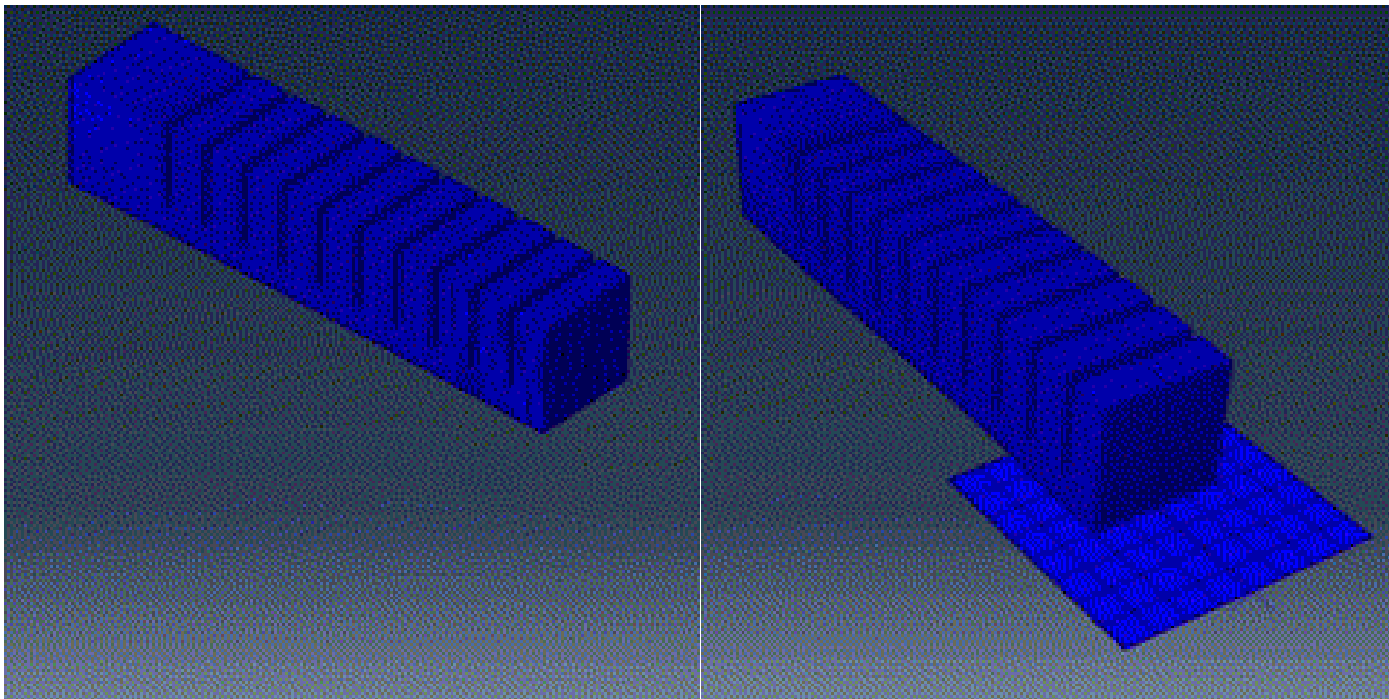




Objectives

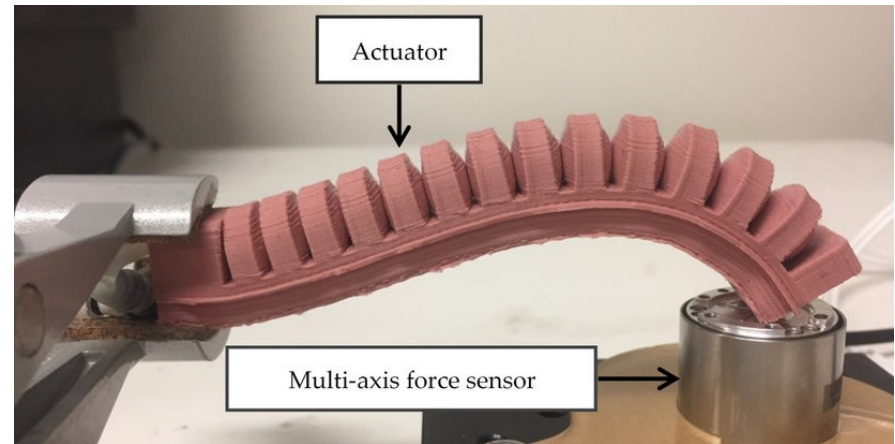
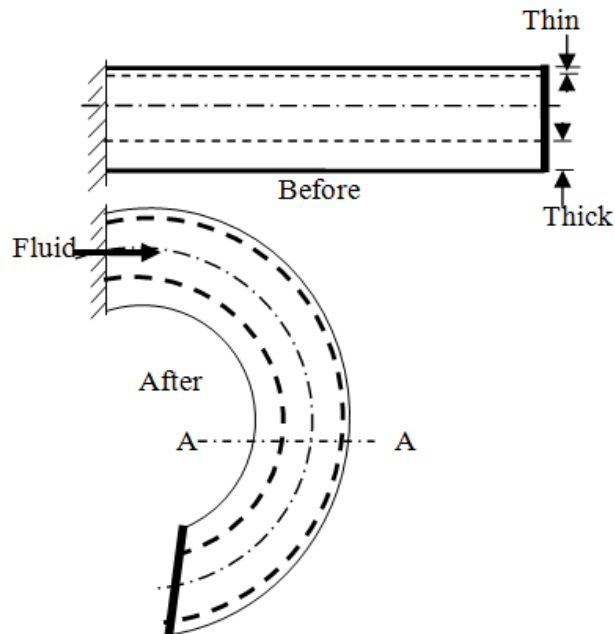
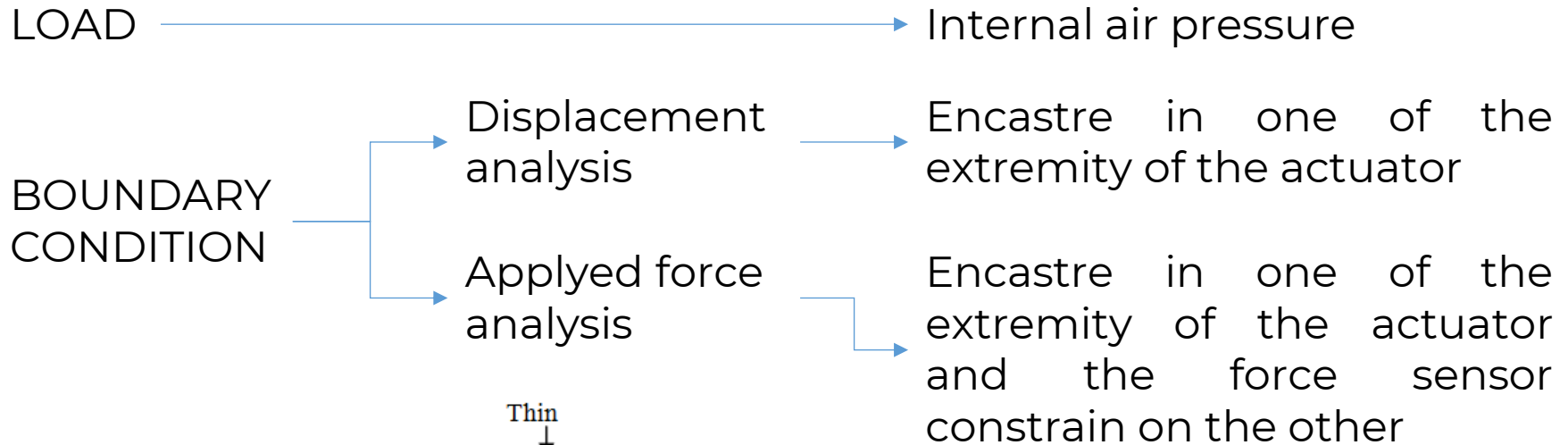
Analysis objectives:

- Evaluate the displacement behaviour of a soft pneumatic actuator depending on the inflating pressure
- Evaluate the applied force by the actuator depending on the inflating pressure



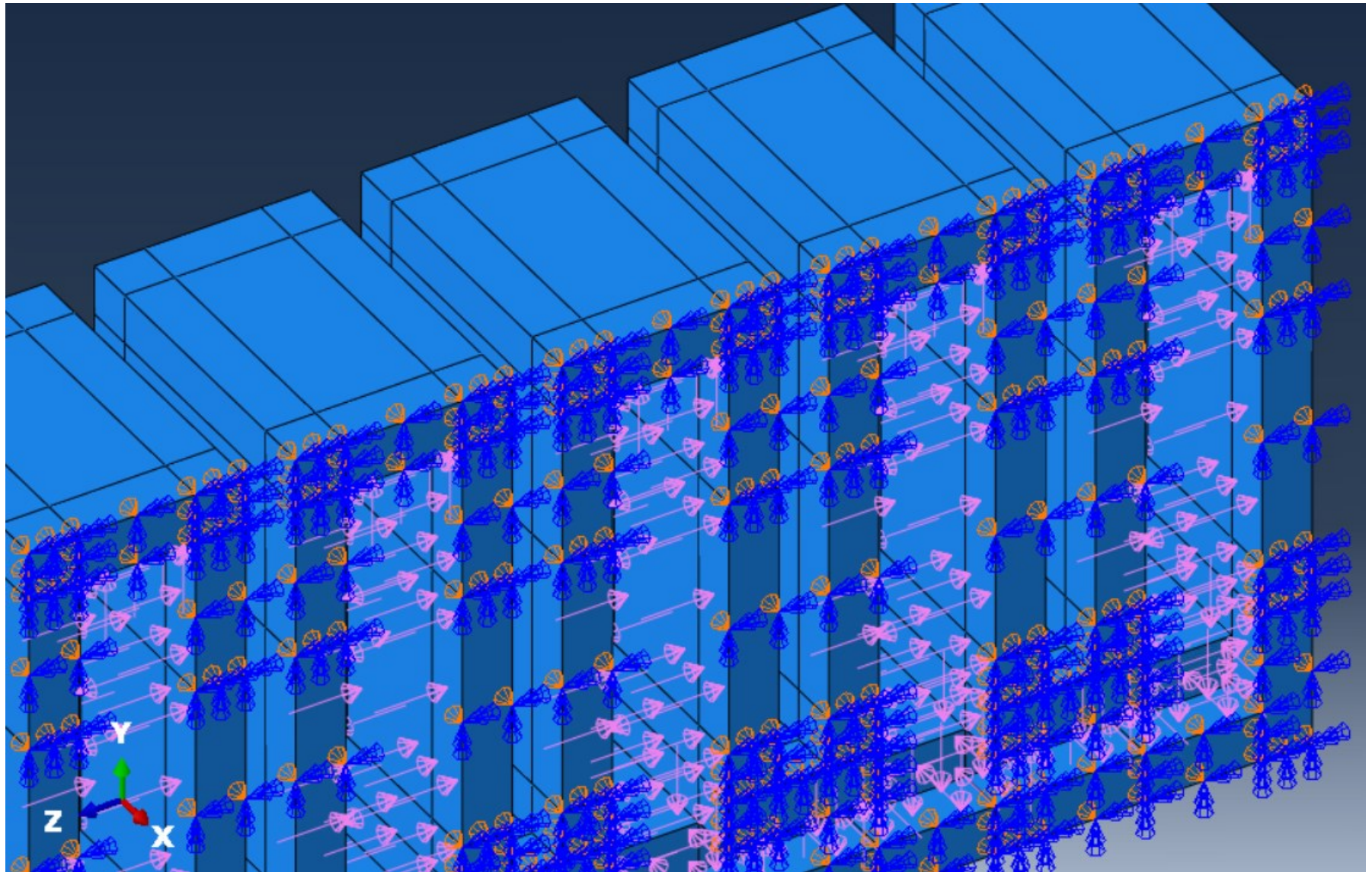


Loading and Boundary Conditions





Loading and Boundary Conditions





Material model

HYPERELASTIC ELASTOMER

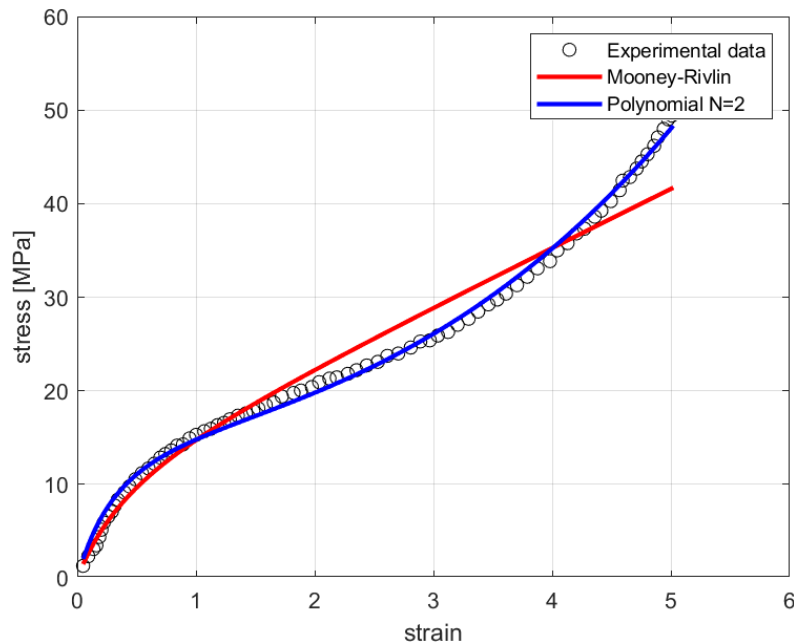
CHARACTERISTICS

- Isotropic and non linear
- Material that exhibith instantaneous elastic response

POLYNOMIAL MODEL

$$W = \sum_{i+j=1}^N C_{ij} (\bar{I}_1 - 3)^i (\bar{I}_2 - 3)^j + \sum_{i=1}^N \frac{1}{D_i} (J^{el} - 1)^{2i}$$

N = 2

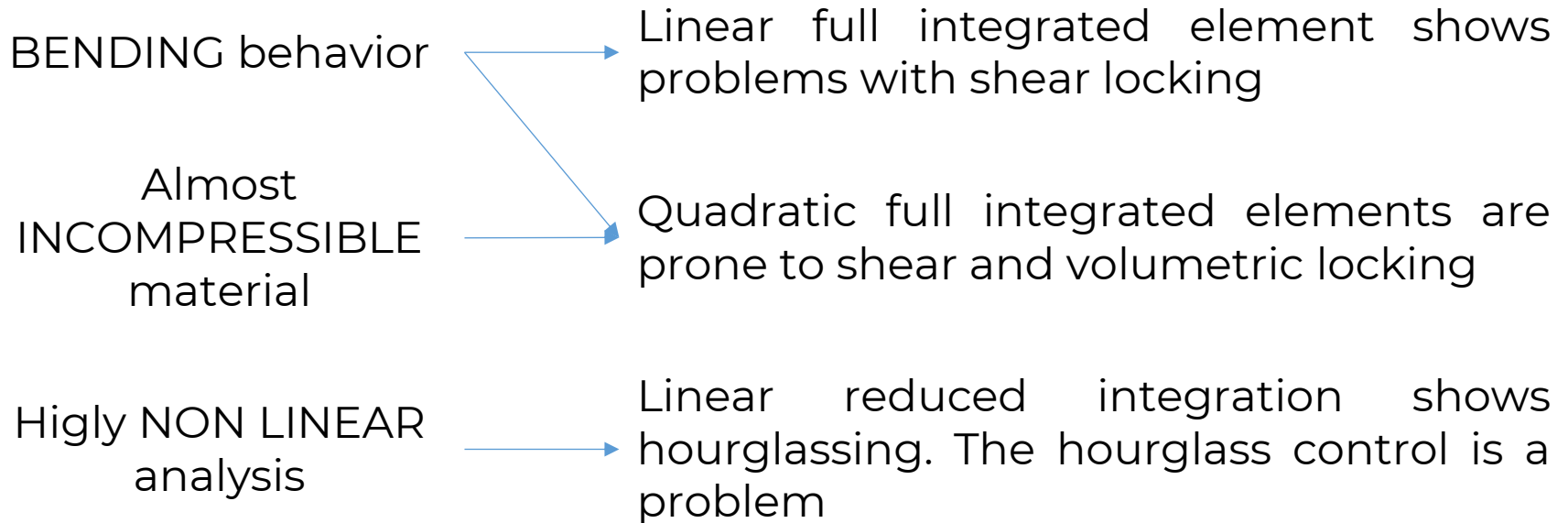


Material parameters

C10	0.55
C01	6.91
C11	7.73E-05
C20	0.032
C02	0.073
D1	0.0029



Element type and dimension



CHOICE → Quadratic reduced integration exahedral element (C3D20R)

PROS

- Faster than full integration
- Less prone to shear and volumetric locking than full Int.

CONS

- Volumetric locking is still possible with large strains
- Localized hourglassing is possible

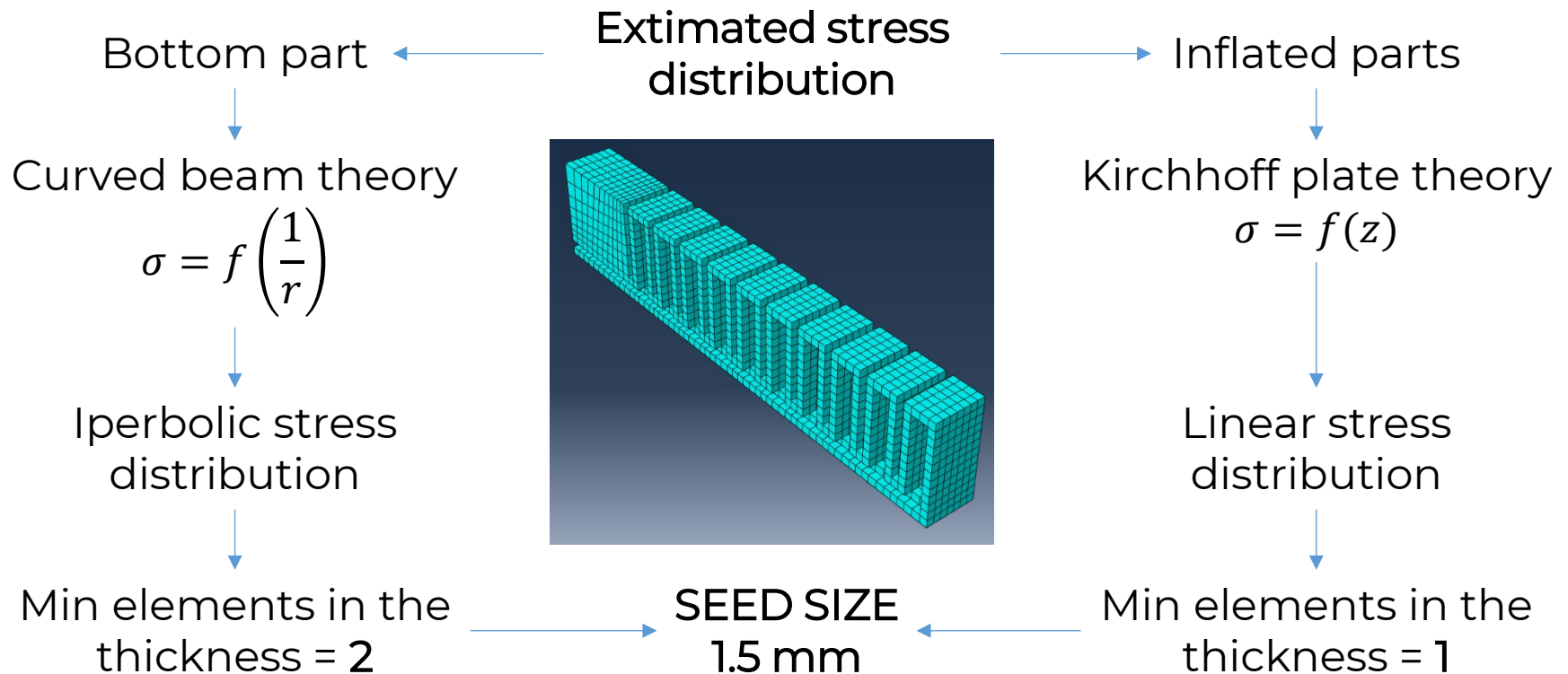


Element type and dimension

Even better choice because of the almost incompressible behaviour could be the hybrid formulation.

Because we didn't treated that in the course I decided to try to find the best solution without it and compare the results.

ELEMENT DIMENSION





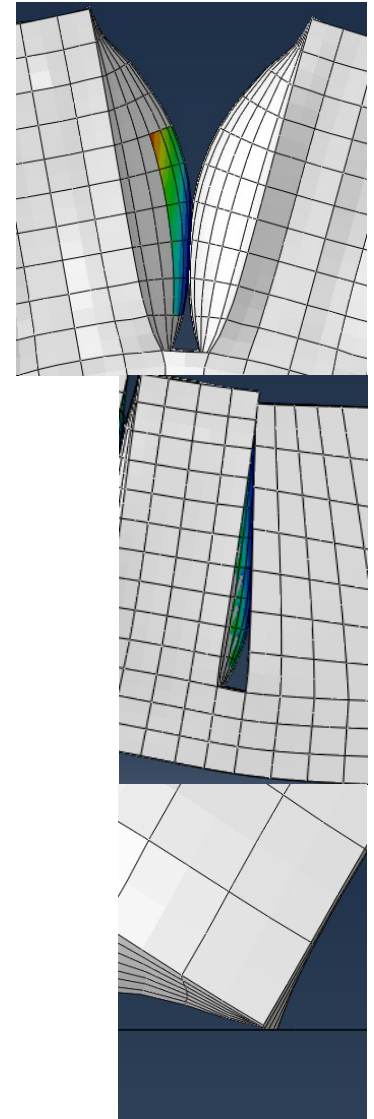
Contact modeling

DISPLACEMENT ANALYSIS

Contact discretization	→	Surface to surface
Contact tracking	→	Small sliding
Contact enforcement	→	Direct

APPLIED FORCE ANALYSIS

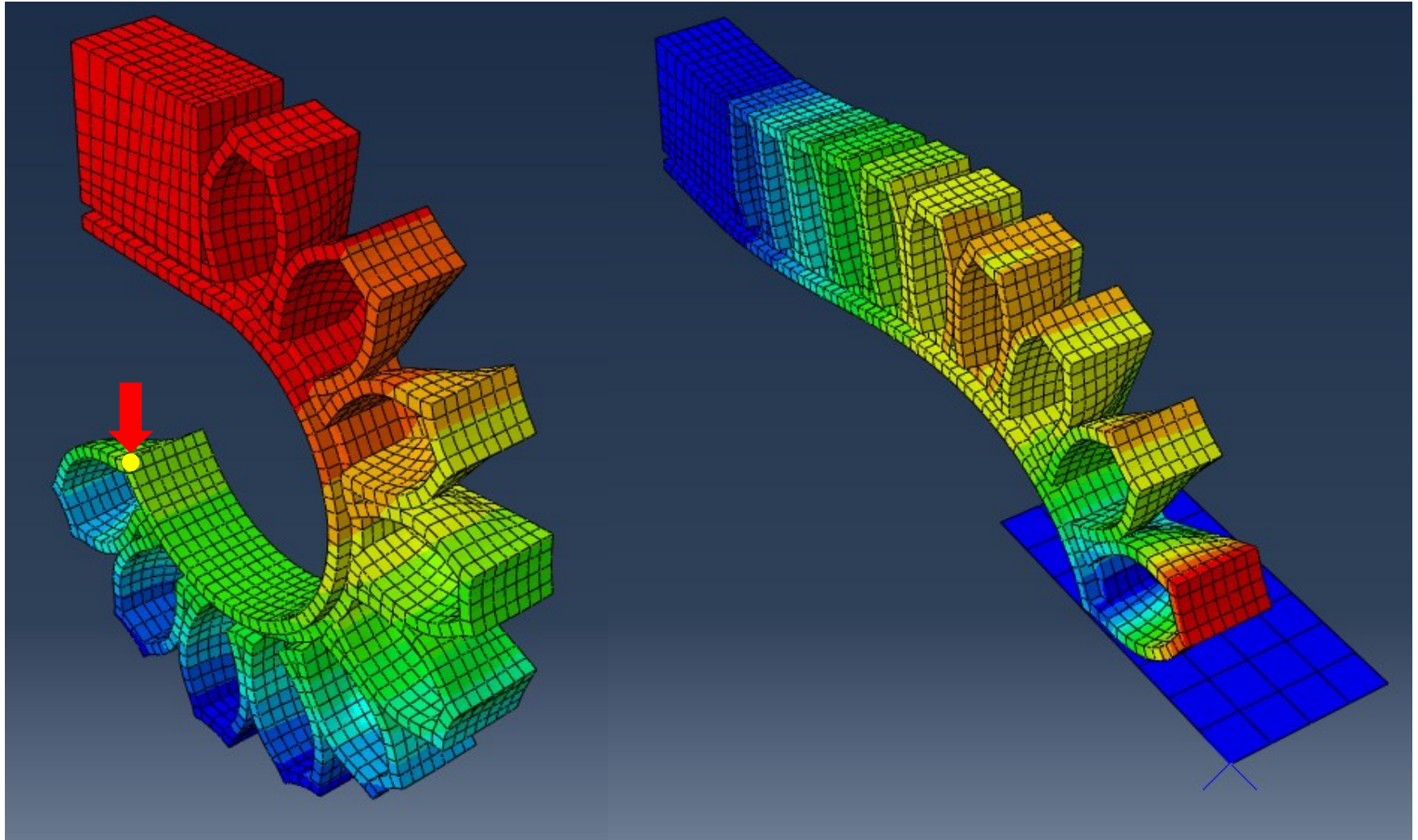
Between softbot surfaces	{	Contact discretization	→	Surface to surface
		Contact tracking	→	Finite sliding
		Contact enforcement	→	Direct
Between softbot and the «floor»	{	Contact discretization	→	Node to surface
		Contact tracking	→	Finite sliding
		Contact enforcement	→	Penalty method





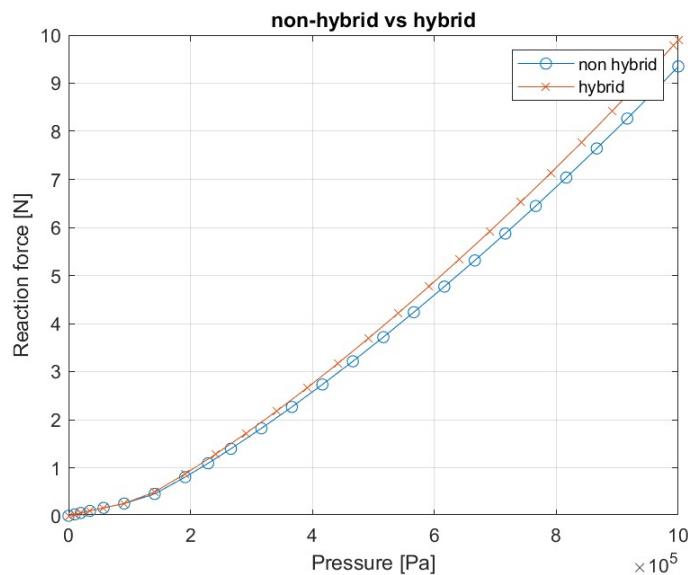
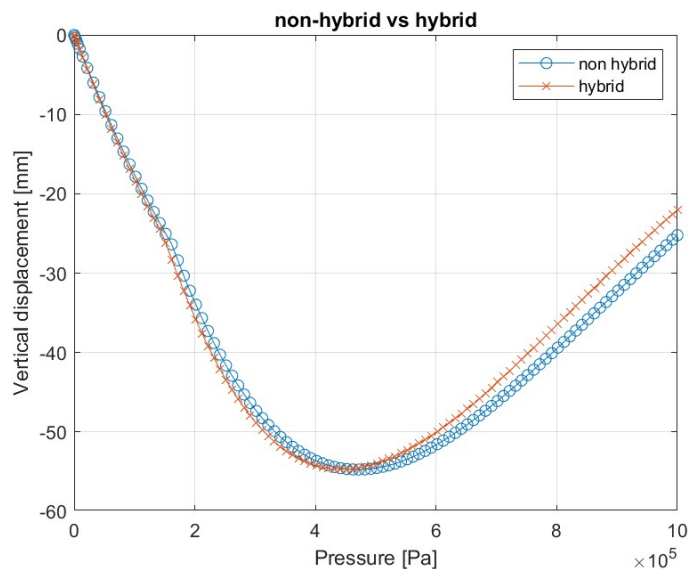
Results

Displacement field





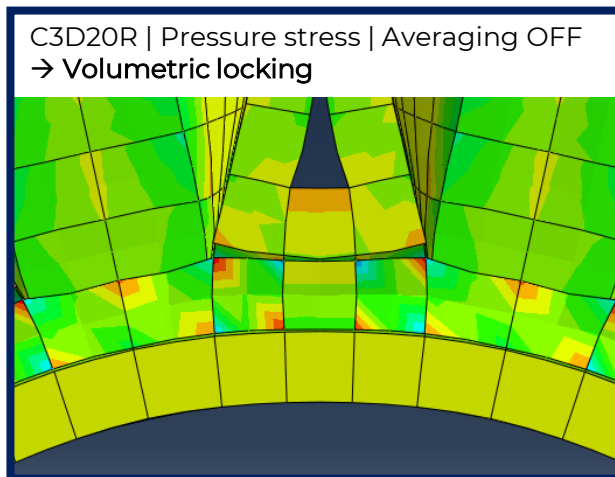
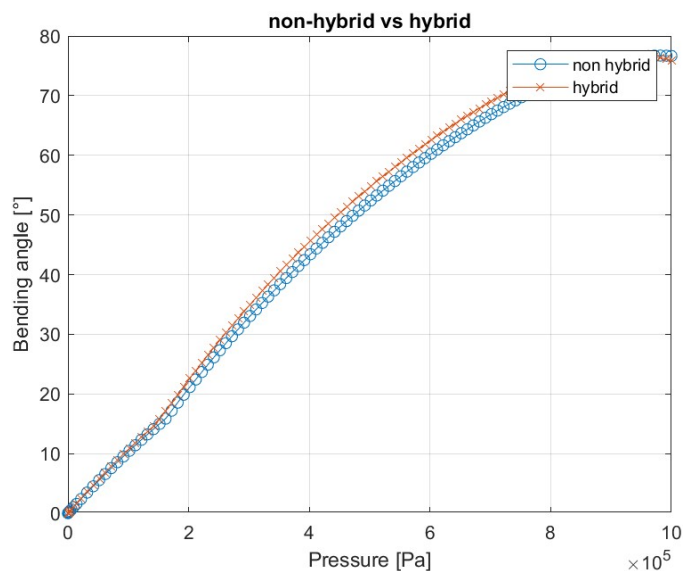
Results



Volumetric locking

Absorbs energy

Model appear to be **stiffer**



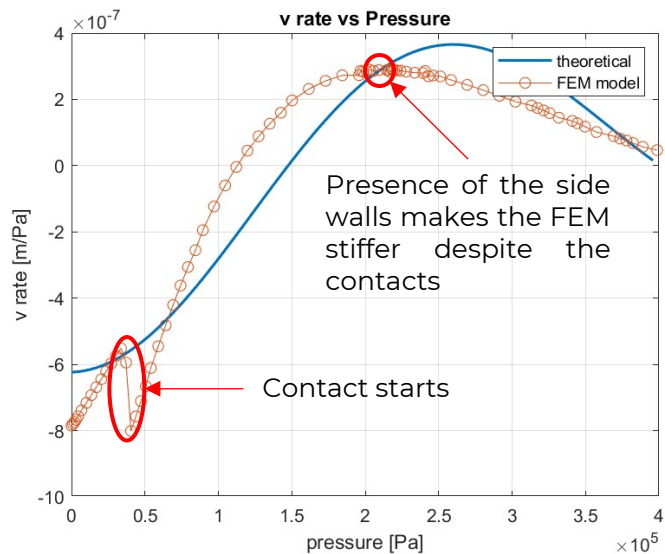
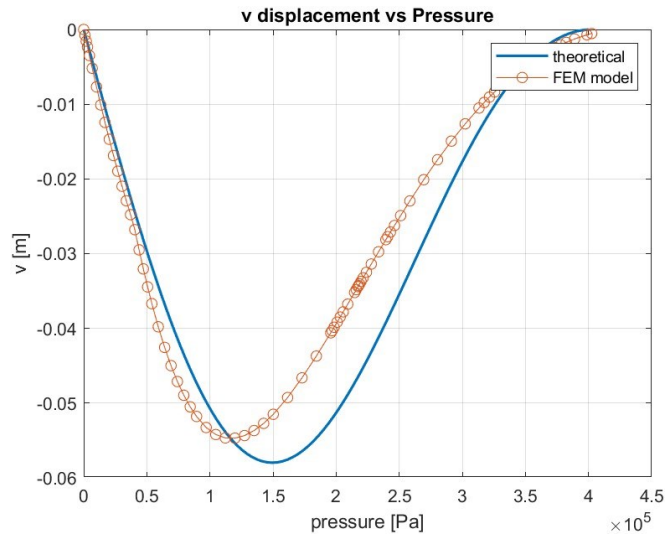
Possible solutions

Refining the mesh

Using the hybrid formulation



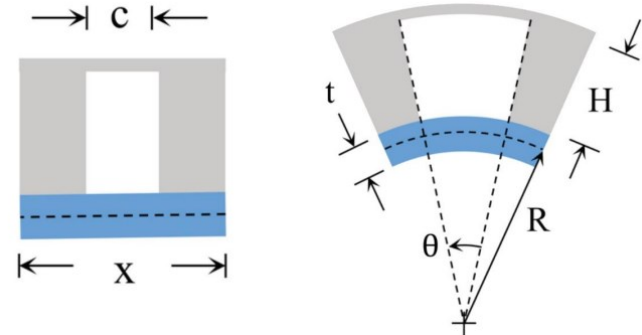
Discussion of results



ANALYTICAL MODEL

$$\frac{1}{R} = \frac{1}{E} \left[\frac{6vq}{t^3} + \frac{6q}{t^3} (x^2 + 2lx + l^2) - \frac{6q(8 + 5v)}{5t} \right]$$

$$q = \frac{PH^2cw}{L^3}$$



For this comparison a linear elastic material was used with both the models.

Analytical characteristics:

- No contact modeled
- No side walls are present

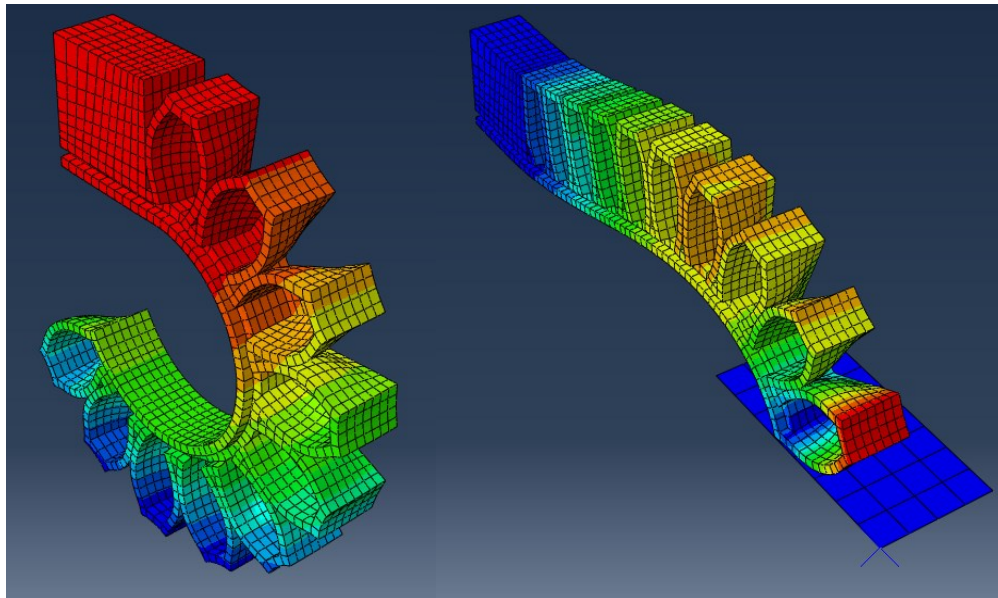


Conclusion

Here are reported briefly the results obtained in the analysis performed.

	C3D20R	C3D20RH
End displ. [mm]	-25.2175	-22.0224
Max RF [N]	9.3498	9.8997

% diff between FEM and analytical models
9%



	Sim times [min]
C3D20R Displacement	13.8
C3D20RH Displacement	10.7
C3D20R Reaction force	9.2
C3D20RH Reaction force	8.8

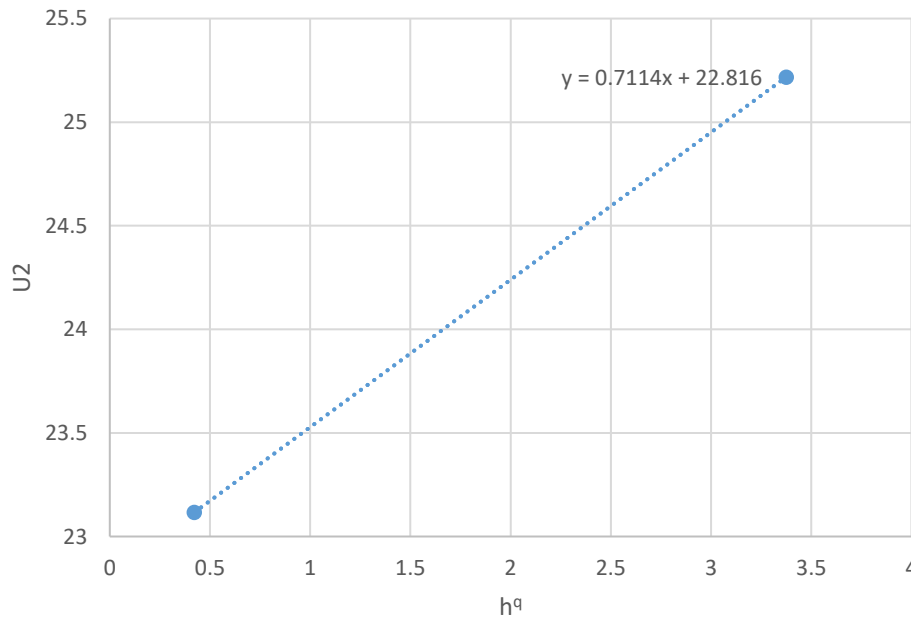


Convergence analysis

Convergence analysis characteristics (for the displacement):

- H refinement
- Only 2 meshes because of the student limitation
- Assumed error order $q = 3$ (quadratic element have been used and the displacement has been evaluated)

Convergence analysis



h	U2	e%
1.5	25.2175	11%
0.75	23.1166	1%

$$U2_{\infty} = 22.816$$

Time taken for the 0.75
mesh dimension = 3 hour